Assignment AB: Two Exclusive Options: A XOR B

Machine Learning for Natural Language Processing 1

Simon Clematide

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Overview

Options
- A
- B

Organization

Assignment AB: Option A: Paper Dissection

Identify an interesting and high-quality (short) NLP paper

- Interesting: **landmark paper** from lecture/reading or ACL
- Interesting: **paper** from or on HF Leaderboards or nlpprogress.com or Papers with Code
- If paper is long and covers many Machine Learning approaches, focus on the best or clearest setup
- Do not select survey papers (you can use them to select a specific paper)

Understand the paper

- Read the paper “quickly and effectively” (next slide) (and reject early if not suitable)
- If you don’t understand some concepts, use supporting resources: presentation videos, blogs. Discuss with ChatGPT & Co., perplexity.ai etc. But make sure, that every sentence that you write makes sense to you!

Typical Paper Structure: IMRaD: Introduction, Methods, Results and Discussion

Efficient reading order may not be the IMRaD order▲

- Abstract
- Conclusion
- Look at examples/figures/tables
- Introduction
- Methods
- Results
- Discussion

Writing Your Paper Dissection: Max. 2 Pages

Important: **Answer these questions in order in your dissection!**

1. What is it about? What problem does it try to solve? Why is it interesting?
2. Which ML methods are used? What is the main innovation of the paper?
3. What are the takeaways?
4. What are possible problems of the approach? Think critically!

Some rules

— What does one need to know for understanding the paper? List the resources that were helpful for you.

— You can also copy/paste the most important figure/table

— You can add a mind map if you like

— Do not just use ChatGPT output!

Option B: Short Student Talk

- 7 minutes + 3 minutes questions (max 7 slides)
- In 2 slots in class, 2 slots in tutorial in December sessions
- Or: create a short screencast (e.g. with Screencastify) for “future” students (no perfectionism asked for!); e.g. a walkthrough to a code example

Topics

- A (short) paper on a technical or social/ethical aspect of ML in NLP
- A technical topic: GPU/TPUs; hierarchical Softmax; feature hashing; different optimizers (Adam); walkthrough of the code of a paper (Papers with Code)

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Organization and Deadlines

For this exercise, you can team up in pairs or work alone.

Communicate your topics and suggestions via Feedback-Forum in OLAT

- For talks: Reply ASAP in forum thread “Student Talks” in OLAT and email me at the same time.
- Paper dissections: Friday 16.1.2026 23:59: Hand-in your PDF in OLAT
- Screencasts: Friday 16.1.2026 23:59: Hand-in Link to screencast in OLAT

List of Important Links I

Slide

1 Background: **Efficient reading** order may not be the IMRaD order ▲ 4